



Modelling Economic Policy Issues



Internet access and household consumption: Impact on expenditure, food diversity, and macronutrient intake in India

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ABSTRACT

Recent progress in internet accessibility have significantly altered household consumption patterns, affecting expenditure decisions, dietary diversity, and macronutrient intake. This study examines how internet accessibility influences household consumption behavior in India using the Household Consumption Expenditure Survey (HCES) 2022–23 survey. To address endogeneity, we employ an Instrumental Variable (IV) framework, complemented by the Plausibly Exogenous method and Coarsened Exact Matching. Results reveal that internet-connected households exhibit higher overall spending, particularly on durable goods, consumables, and food, alongside greater discretionary expenditure. Internet access also enhances dietary outcomes, with connected households consuming more calories, macronutrients, and a wider variety of foods. The findings underscore the need for policies that expand affordable internet, promote digital literacy, and integrate online platforms into nutrition and financial education, thereby ensuring inclusive welfare gains from digitalization.

1. Introduction

The rapid expansion of internet access has fundamentally reshaped multiple dimensions of human life, including education, healthcare, financial decision-making, and household consumption. Early evidence highlights that internet growth has altered demand patterns, particularly for entertainment and lifestyle goods (Hong, 2007). More recent studies demonstrate that digital connectivity extends well beyond leisure, influencing household expenditure, savings, and well-being across diverse contexts (Luo et al., 2019; Mora-Rivera and García-Mora, 2021; Thariqi et al., 2021).

Household welfare is not determined by income alone, but also by the quality and composition of consumption (Noll and Weick, 2015). Internet access increasingly shapes these consumption choices by enabling access to e-commerce platforms, online grocery services, and digital financial tools (Chunfang et al., 2023; Wu et al., 2024). Empirical evidence from China, Indonesia, and Mexico suggests that internet penetration raises overall expenditure, particularly on food, durable goods, and services, while also enhancing consumption diversity (Gurning and Khaliqi, 2019; García-Mora and Mora-Rivera, 2021; Vatsa et al., 2022).

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Dietary outcomes are another critical channel through which digitalization affects household well-being. Studies in rural China indicate that internet access improves caloric intake, macronutrient consumption, and diet quality (Xue et al., 2021; Ma and Jin, 2022; Cui et al., 2024). Similarly, internet usage facilitates exposure to diverse cuisines and healthier food choices, broadening household food baskets and improving nutritional balance (Vatsa et al., 2022; Liu et al., 2025). At the same time, concerns have been raised regarding the influence of targeted digital advertising, particularly the promotion of calorie-dense processed foods, which may offset some of these potential nutritional gains (Benvenuto et al., 2022; Gao et al., 2023).

Beyond food, digital connectivity also influences health expenditures and mental well-being, underscoring its broader socio-economic impact. For instance, internet usage has been linked to improved healthcare consumption and dietary awareness (Benvenuto et al., 2022; Yang, 2021), while also demonstrating heterogeneous effects on mental health outcomes, especially among older populations (Zhu et al., 2023). In rural settings, internet access is further associated with reductions in poverty and improvements in resource use, such as cooking energy and electricity consumption (Mora-Rivera and García-Mora, 2021; Zhang et al., 2023; Gao et al., 2023). Despite this growing body of global evidence, research on the relationship between internet connectivity, household expenditure, and nutritional intake in India remains limited.

This study is an attempt to fill this gap in the literature by examining the impact of Internet access on household consumption behaviour in India while focussing on impact on expenditure, food diversity, and macronutrient intake. The study uses nationally representative Household Consumption Expenditure Survey (HCES) 2022–23 to further examine the relationship between digitally inclusivity and household welfare outcomes. To ensure robust causal inference, the study employs instrument variable estimation, the plausibly exogenous methods and matching (CEM) to address potential endogeneity in internet usage decisions. Remaining of the paper is organised in such a way that the next section provides a review of the relevant literature, situating the study within existing scholarship landscape and identifies key research gaps, which is further followed by a detailed description of the data sources and methodological approach employed to do the study. The subsequent section presents the main findings of the analysis. Finally, the article concludes with a discussion of the results and their policy implications, drawing connections to prior research and outlining directions for future inquiry.

2. Literature review

Although access to the internet and household consumption has become an important area for investigation for scholars today, the scholarship in this area has predominantly evolved in recent years only (Parappurathu, 2015; Gunawan, 2022). Scholars started studying this very important relationship only in late 2000s early 2010s which also aligns with the internet penetration in most parts of the world (Zhang et al., 2020; Wang and Hao, 2018; Kellerman, 2004; Zhang and Meng, 2019; Liu et al., 2025; Chinn and Fairlie, 2010; Imansyah, 2018; Lewis, 2018; Leung et al., 2025). After moving from how scholarship started emerging in the domain, this section looks at review of existing work in the Indian context followed by the international context. Finally, the review of the literature section also identifies the gaps and thus paving the way for the need for conducting this research. With internet penetration rising sharply and consumption patterns evolving rapidly, India provides a critical setting to investigate these dynamics.

In Indian Context, access to the internet expanded gradually from the early 2000s and reached about 10 percent by 2011, rigorous scholarly work was also absent around the same time. It is only around 2016 where the people in the country experienced substantial expansion of smart phone based affordable internet services, and also pushed internet penetration to over half of the population by 2021. This expansion also led to the emergence of research studying the relation between digital connectivity shaping people's consumption behaviors and the relationship got all the more pronounced during pandemic lockdown (Gavaravarapu et al., 2022). Gavaravarapu et al. (2022) found that searches for immunity boosters, nutritional supplements, vitamin supplements, and indigenous remedies among Indian internet users increased during the pandemic.

The study also suggested that most users were dependent on the social media for the macro nutritional guidance, while the ones affected with covid-19 consulted health professionals. Impact of internet exposure was not limited to the nutritional or health aspect, access to internet significantly contributed to the proliferation of both regional as well as international cuisines in the urban spaces (Kumar et al., 2022; Rathi et al., 2025). However, in rural India, food and nutrition are largely affected by market access and women's bargaining power at home rather than just internet access (Gupta et al., 2020; Sangwan and Kumar, 2021).

Furthering the scholarship landscape to Asia, scholars suggest similar insights with some specific regional variations. Research from the Chinese context shows that access to the internet significantly improves dietary quality and diversity not just by increasing access to dietary knowledge but also because of enhanced household income opportunities (Cui et al., 2024; Ma and Jin, 2022; Deng et al., 2024; Yang et al., 2023; Xiang and Zhang, 2025). Just like China, Indonesia has been of great interest among scholars for these kinds of studies because of various reasons such as early rapid expansion of the internet in comparison to other developing nations, strong e-commerce ecosystem, focused government schemes enabling natural experiments for social scientists and availability of rich household level data (Sentosa et al., 2025; Rahman et al., 2025). Studies show variety of shades in the findings from Indonesian context, while internet connectivity leads to enhanced consumption expenditure and in most cases nutrition too due to increased information and income opportunities, it in many cases also reduce healthy food consumption especially in lower socio-economic strata, hinting towards role of socio-economic factors in mediating these effects (Mahmudiono et al., 2022; Martha et al., 2021).

It is evident from studies from China and Indonesia, that the global scenario about the influence of internet connectivity on household consumption and finer composition of it is multifaceted. A systematic review study indicates the internet ecosystem as an intervention increases the chances for healthy eating behaviors across different population groups, particularly among the ones suffering from chronic illnesses (Seid et al., 2024). However, alarming rates of obesity and non-communicable diseases worldwide due to the increased consumption of sugar, fats and processed foods show the other side of the question in context here (Dai et al., 2022).

The world is going through a nutrition transition which is accelerated by the evolving digital ecosystem (Millar et al., 2025).

Existing work is largely concentrated in China (Luo et al., 2019; Xue et al., 2021; Cui et al., 2024) and selected emerging economies such as Mexico and Indonesia (Mora-Rivera and García-Mora, 2021; Thaqiri et al., 2021), leaving the Indian context underexplored. With internet penetration rising sharply and consumption patterns evolving rapidly, India provides a critical setting to investigate these dynamics.

This study seeks to fill this void in scholarship in this field by examining how internet access influences household expenditure, food diversity, and macronutrient intake in India using data from the Household Consumption Expenditure Survey (HCES) 2022–23 (MoSPI, 2023).

3. Data

This study uses unit-level data from the Household Consumption Expenditure Survey (HCES) conducted by the National Statistical Office (NSO) during August 2022–July 2023. The HCES is a nationally representative survey designed to collect information on household consumption of goods and services, providing critical inputs for measuring living standards, poverty, and consumer price indices. The survey follows a multi-stage stratified sampling design, covering both rural and urban areas across all States and Union Territories of India, with the exception of some inaccessible regions of the Andaman and Nicobar Islands.

The HCES 2022–23 collected data from 2,61,953 households (1,54,357 rural and 1,07,596 urban), drawn from 15,016 First Stage Units (FSUs)—8723 in rural and 6115 in urban areas. The FSUs were selected using Simple Random Sampling Without Replacement (SRSWOR), and households served as the Ultimate Stage Units (USUs). To minimize bias from questionnaire sequencing, six possible orders of administration were implemented across households.

The survey employed four instruments: the Food Expenditure Questionnaire (FDQ), the Consumables and Services Questionnaire (CSQ), the Durable Goods Questionnaire (DGQ), and the Household Characteristics Questionnaire (HCQ). Each household was visited thrice within a quarter to record detailed expenditures on food, non-food consumables, services, and durable goods, along with demographic and socioeconomic characteristics.

Data collection was organized into 10 rotating panels, each spanning three months between August 2022 and July 2023, ensuring temporal balance in expenditure reporting. Given the absence of direct income measures, a wealth index was constructed using

Table 1
Descriptive statistics.

Variable	N	Mean	Std. Dev.	Min	Max
Internet Availability	261,739	0.692	0.461	0	1
Leave-One-Out Proportion of Internet Users in FSU (IV)	261,729	0.692	0.273	0	1
Total Expenditure	258,268	62,558.37	80,067.58	1021	3663,504
Food Expenditure	261,075	3475.47	2899.96	10	200,290
Consumables Expenditure	261,257	25,319.42	44,516.07	86	3258,958
Durables Expenditure	258,995	33,564.44	57,014.48	120	3242,500
Total Daily Calories	257,936	6837.27	5153.98	34.71	583,359
Total Daily Proteins	258,036	116.09	77.02	0	2833
Total Daily Carbohydrates	261,075	950.78	533.40	0	20,689
Total Daily Fats	261,075	171.61	124.02	0	24,792
Age (years)	261,739	48.63	14.30	13	117
Male Household Head (Dummy)	261,739	0.854	0.353	0	1
Household Size	261,739	4.29	2.11	1	37
No Schooling (Dummy)	261,739	0.235	0.424	0	1
Up to Primary (Dummy)	261,739	0.235	0.424	0	1
Secondary Education (Dummy)	261,739	0.283	0.450	0	1
Higher Secondary/Diploma (Dummy)	261,739	0.114	0.318	0	1
Graduate and Above (Dummy)	261,739	0.133	0.340	0	1
Poorest Quintile (Dummy)	261,739	0.248	0.432	0	1
Poorer Quintile (Dummy)	261,739	0.160	0.367	0	1
Middle Quintile (Dummy)	261,739	0.192	0.394	0	1
Richer Quintile (Dummy)	261,739	0.204	0.403	0	1
Richest Quintile (Dummy)	261,739	0.195	0.397	0	1
Hindu (Dummy)	261,739	0.771	0.420	0	1
Muslim (Dummy)	261,739	0.122	0.328	0	1
Christian (Dummy)	261,739	0.068	0.251	0	1
Sikh (Dummy)	261,739	0.015	0.123	0	1
Jain (Dummy)	261,739	0.002	0.046	0	1
Buddhist (Dummy)	261,739	0.013	0.114	0	1
Other Religions (Dummy)	261,739	0.009	0.093	0	1
Scheduled Caste (SC) (Dummy) ¹	261,739	0.170	0.376	0	1
Scheduled Tribe (ST) (Dummy)	261,739	0.146	0.353	0	1
Other Backward Class (OBC, Dummy)	261,739	0.399	0.490	0	1
Rural (Dummy)	261,739	0.592	0.491	0	1

Source: Author's Calculation.

Variable definitions are reported in [Appendix A](#).

Principal Component Analysis (PCA) as a proxy for household income.

This dataset is particularly suitable for empirical analysis as it provides granular information on household expenditure patterns, food consumption, and durable goods ownership. These features allow us to examine the impact of Internet connectivity on consumption behaviour and dietary diversity while controlling for household characteristics. The analysis uses data from the Household Consumption Expenditure Survey (HCES) 2022–23, which covers 261,953 households (154,357 rural and 107,596 urban) across India. Since the objective of the study is to examine consumption behavior at the household level, the raw data—originally recorded at the item level—were aggregated to construct a single observation per household.

Specifically, expenditures on individual items were aggregated to compute total food and non-food expenditure (including durables and services) at the household level. Similarly, nutritional intake variables—calories, proteins, fats, and carbohydrates—were computed by combining reported quantities of food consumption with standard nutritional conversion factors (based on USDA food composition data), and then aggregated to the household level.

As part of the data cleaning process, observations with missing data were excluded. These cases constituted a very small fraction of the overall sample and did not materially affect the sample size or its representativeness.

Table 1 presents descriptive statistics for all variables. We begin with our independent variable of interest - internet availability at the household level - followed by our instrument, the leave-one-out proportion of internet users in the FSU.

Next, we present the dependent variables measuring expenditure and dietary items, all expressed in levels.

Finally, we report the full set of control variables included across all specifications. Since the majority of these controls are dummy variables - with the exception of age and household size - one category is omitted for each categorical variable (education, wealth index, religion, and caste) to avoid multicollinearity from perfect collinearity among the dummies.

The descriptive statistics indicate that approximately 69.2% of households in the sample have access to the internet, suggesting relatively widespread but not universal connectivity. The instrumental variable - measured as the leave-one-out proportion of internet users within the FSU closely mirrors this average, indicating consistency between individual adoption and local diffusion patterns.

Household expenditure exhibits substantial variation, with mean total expenditure of ₹62,558 and a large standard deviation, reflecting significant heterogeneity in consumption levels across households. Similar dispersion is observed across food, consumables, and durables expenditure, pointing to unequal spending patterns. Nutritional intake variables, including calories, proteins, carbohydrates, and fats, also display wide ranges, suggesting disparities in diet quality and quantity.

The average age of the household head is approximately 49 years, and a large majority (85.4%) of households are male-headed. The mean household size is about 4.3 members. Educational attainment is fairly distributed across categories, though a notable proportion of households have low levels of schooling.

The sample is predominantly Hindu (77.1%), followed by Muslims and Christians, reflecting the broader demographic composition. Social group distribution shows substantial representation of Other Backward Classes (OBCs), along with Scheduled Castes (SCs) and Scheduled Tribes (STs). Finally, about 59.2% of households reside in rural areas, indicating a rural majority.

Overall, the summary statistics highlight considerable heterogeneity across households in terms of economic status, demographic characteristics, and consumption patterns, underscoring the importance of controlling for these factors in the empirical analysis.

4. Empirical strategy

We estimate the effect of household-level internet access on several expenditure measures expressed in logarithms,¹ enabling recovery of the average relative effect of connectivity on consumption. The estimating equation is:

$$\text{Expenditure}_i = X_i \beta + \text{Internet}_i \gamma + \alpha_i + \varepsilon_i$$

where Expenditure_i denotes a measure of household expenditure (or, in some specifications, a measure of diet quality and composition), X_i is a vector of household-level controls, and Internet_i is an indicator for household internet access - our variable of interest. Location fixed effects α_i are defined at the primary sampling unit (FSU) or district level, depending on the specification. All estimates are weighted to account for the stratified, multi-stage sampling design of the Household Consumption and Expenditure Survey, ensuring national representativity. Standard errors are clustered at the FSU level to account for within-cluster error correlation induced by the survey design.

4.1. Endogeneity and identification

Since internet access is a choice variable, it is likely endogenous: households that select into connectivity may differ systematically from non-connected households along dimensions that also affect expenditure. We confirm the presence of endogeneity using a heteroskedasticity-robust Hausman test, implemented via the control function approach of Wooldridge (2015).

We address this concern using two complementary strategies.

Two-Stage Least Squares. The first stage takes the form:

¹ Since the dependent variable is in logs, the coefficient γ can be interpreted as the percentage change in expenditure associated with internet access.

$$Internet_i = X_i \beta + Z_i \gamma + \alpha_i + \nu_i$$

where Z_i - our instrument - is the *Leave-one-out internet adoption rate at the FSU level*, computed as:

$$IV = \frac{\text{Number of Internet - connected households in FSU} - 1(\text{household}_i \text{ has internet})}{(\text{Total number of households in FSU}) - 1}$$

This instrument captures the proportion of *other* households within the same FSU that have internet access, excluding the household of interest to avoid mechanical correlation with the dependent variable.

Because within-FSU variation in this instrument is limited, identification relies on district fixed effects; we verify this is appropriate by showing that OLS estimates are stable across FSU and district fixed-effect specifications, confirming that district fixed effects absorb sufficient unobserved heterogeneity.

The exclusion restriction requires $E(Z_i \varepsilon_i) = 0$, which amounts to imposing $\gamma = 0$ in the reduced-form equation:

$$Expenditure_i = X_i \beta + Z_i \gamma + \alpha_i + \varepsilon_i$$

That is, Z_i must affect expenditure only through internet adoption, with no direct channel. To assess robustness to potential violations of this restriction, we apply the **Plausibly Exogenous** approach of Conley et al. (2012), which constructs valid confidence intervals under the relaxation that $\gamma \neq 0$. Concretely, we consider the family of shifted regressions:

$$(Expenditure_i - Z_i \gamma_0) = X_i \beta + \alpha_i + \varepsilon_i \quad \text{for } \gamma_0 \in [\gamma_{\min}; \gamma_{\max}]$$

guided by the reduced-form coefficient. Across all expenditure outcomes, we allow the direct effect of the instrument to be as large as $\pm 50\%$ of the reduced-form estimate, providing a very demanding test of the robustness of our 2SLS results.

Coarsened Exact Matching. As a second identification strategy, we apply the Coarsened Exact Matching (CEM) algorithm of Iacus et al. (2011), which works by temporarily "coarsening" variables (e.g., binning continuous variables) to match units that are similar enough, then discarding unmatched units. It reduces model dependence by pre-processing the data to achieve covariate balance between connected and unconnected households. Valid causal inference under matching requires two conditions:

1. **Conditional independence (CIA):** conditional on observed covariates, treatment assignment is independent of potential outcomes,

$$(Y_1, Y_0) \perp D \mid X$$

2. **Common support:** the probability of treatment is bounded away from 0 to 1 within each stratum,
 $0 < Pr(D = 1 \mid X) < 1$ almost surely

Table 2
Effect of internet access on household expenditure (Log).

	Ln of Total Expenditure	Ln of Food Expenditure	Ln of Consumables Expenditure	Ln of Durables Expenditure
Panel A: OLS				
β FSU FE	.244***	.124***	.267***	.249***
(SE)	(0.005)	(0.003)	(0.006)	(0.005)
R2	.641	.68	.556	.605
β District FE	.232***	.115***	.256***	.237***
(SE)	(0.006)	(0.004)	(0.007)	(0.006)
R2	.497	.477	.41	.42
N	261,739	261,075	261,257	258,995
Panel B: Control function: District FE				
Beta	.262***	.142***	.408***	.212***
(Bootstrap SE)	(0.018)	(0.014)	(0.022)	(0.019)
1st st residuals	−0.039**	−0.034**	−0.192***	.032*
(Bootstrap SE)	(0.018)	(0.013)	(0.022)	(0.019)
N	261,729	261,065	261,247	258,985
R2	.497	.477	.411	.42
Panel C: 2SLS: District FE				
Beta	.262***	.142***	.408***	.212***
(SE)	(0.018)	(0.014)	(0.022)	(0.019)
N	261,729	261,065	261,247	258,985
R2	.483	.463	.388	.407
Panel D: Reduced form: District FE				
Beta	.204***	.11***	.317***	.165***
(SE)	(0.014)	(0.011)	(0.017)	(0.015)
N	261,729	261,065	261,247	258,985
R2	.488	.473	.406	.409
Panel E: Plausibly exogenous: UCI: District FE				
95% CI of β	[.228; 0.426]	[.116; 0.246]	[.366; 0.658]	[.175; 0.352]
N	261,729	261,065	261,247	258,985

Source: Author's Calculation.

5. Empirical Analysis

Table 2 presents our main results for the coefficient of interest - internet availability - across instrumented and non-instrumented specifications.

1) Control Function Approach (Panel B): We next instrument the endogenous variable. Panel B reports the control function approach, in which the first-stage residuals are included in the main equation. Following Wooldridge (2015), this yields a heteroskedasticity-robust Hausman test of endogeneity. Standard errors are corrected for the inclusion of a generated regressor using 999 replications of the wild cluster bootstrap. The first-stage residuals are statistically significant across all four columns, providing clear justification for the instrumental variables strategy.

The first-stage coefficient of our IV is positive and highly significant at 0.777 (SE = 0.0045). The Kleibergen-Paap Wald rk F statistic

Table 3

Balance of covariates before and after coarsened exact matching.

Variable	Before matching			After matching		
	(1)	(2)	(3)	(4)	(5)	(6)
	Control	Treated	Difference	Control	Treated	Difference
Age (in Years)	49.022 (15.445)	47.512 (13.857)	−1.511*** (0.060)	47.351 (13.071)	47.337 (13.063)	−0.015 (0.058)
Male Household Head	0.817 (0.387)	0.879 (0.326)	0.063*** (0.001)	0.904 (0.295)	0.904 (0.295)	0.000 (0.001)
Household Size	3.810 (2.168)	4.497 (2.068)	0.687*** (0.009)	4.244 (1.804)	4.289 (1.737)	0.045*** (0.008)
No Schooling	0.388 (0.487)	0.208 (0.406)	−0.180*** (0.002)	0.183 (0.387)	0.183 (0.387)	−0.000 (0.002)
Primary Below	0.255 (0.436)	0.230 (0.421)	−0.025*** (0.002)	0.225 (0.418)	0.225 (0.418)	0.000 (0.002)
Secondary	0.230 (0.421)	0.294 (0.456)	0.065*** (0.002)	0.312 (0.463)	0.312 (0.463)	0.000 (0.002)
Higher Sec. Diploma	0.072 (0.258)	0.123 (0.328)	0.051*** (0.001)	0.121 (0.327)	0.121 (0.327)	0.000 (0.001)
Graduate Above	0.056 (0.229)	0.145 (0.352)	0.089*** (0.001)	0.159 (0.365)	0.159 (0.365)	−0.000 (0.002)
Poorest	0.452 (0.498)	0.180 (0.384)	−0.272*** (0.002)	0.187 (0.390)	0.187 (0.390)	0.000 (0.002)
Poorer	0.198 (0.399)	0.151 (0.358)	−0.048*** (0.002)	0.149 (0.356)	0.149 (0.356)	−0.000 (0.002)
Middle	0.161 (0.368)	0.213 (0.409)	0.052*** (0.002)	0.207 (0.405)	0.207 (0.405)	−0.000 (0.002)
Richer	0.118 (0.322)	0.241 (0.428)	0.124*** (0.002)	0.234 (0.424)	0.234 (0.424)	−0.000 (0.002)
Richest	0.071 (0.257)	0.215 (0.411)	0.144*** (0.002)	0.224 (0.417)	0.224 (0.417)	−0.000 (0.002)
Hinduism	0.845 (0.361)	0.818 (0.386)	−0.027*** (0.002)	0.827 (0.378)	0.827 (0.378)	−0.000 (0.002)
Islam	0.111 (0.314)	0.123 (0.329)	0.013*** (0.001)	0.107 (0.309)	0.107 (0.309)	0.000 (0.001)
Christianity	0.025 (0.157)	0.028 (0.165)	0.003*** (0.001)	0.052 (0.222)	0.052 (0.222)	−0.000 (0.001)
Sikhism	0.008 (0.090)	0.020 (0.139)	0.012*** (0.001)	0.007 (0.085)	0.007 (0.085)	−0.000 (0.000)
Jainism	0.001 (0.026)	0.002 (0.047)	0.002*** (0.000)	0.000 (0.021)	0.000 (0.021)	−0.000 (0.000)
Buddhism	0.007 (0.082)	0.007 (0.082)	0.000 (0.000)	0.004 (0.066)	0.004 (0.066)	−0.000 (0.000)
Other Religions	0.003 (0.054)	0.002 (0.044)	−0.001*** (0.000)	0.002 (0.046)	0.002 (0.046)	−0.000 (0.000)
SC	0.234 (0.424)	0.176 (0.380)	−0.059*** (0.002)	0.154 (0.361)	0.154 (0.361)	−0.000 (0.002)
ST	0.128 (0.334)	0.081 (0.273)	−0.046*** (0.001)	0.114 (0.318)	0.114 (0.318)	−0.000 (0.001)
OBC	0.421 (0.494)	0.449 (0.497)	0.028*** (0.002)	0.425 (0.494)	0.425 (0.494)	−0.000 (0.002)
Rural	0.556 (0.497)	0.579 (0.494)	0.023*** (0.002)	0.610 (0.488)	0.610 (0.488)	0.000 (0.002)
Observations	80,489	181,250	261,739	74,476	153,220	227,696

Source: Author's Calculation.

of 30,173.15 far exceeds the Stock-Yogo maximal critical value of 16.38, and the Anderson-Rubin Wald test yields $\chi^2(1) = 227.59$ ($p \approx 0$). Together, these results conclusively rule out weak instrument concerns.

2) 2SLS Estimates (Panel C): As expected, the 2SLS coefficients are numerically identical to those from the control function approach. The instrumented coefficients are larger in magnitude than their OLS counterparts for all outcomes except durables expenditure. This pattern is consistent with a Local Average Treatment Effect (LATE) interpretation: 2SLS identifies the effect for compliers - households whose internet access is determined by the instrument¹ - who may be precisely those for whom connectivity most meaningfully expands consumption opportunities, through e-commerce access, improved labor market information, or digital financial services. The effect among this subgroup can therefore exceed the average treatment effect (ATE) approximated by OLS.

3) Reduced Form (Panel D): Panel D reports reduced form estimates, replacing the endogenous variable with the instrument in the main equation. This serves two purposes: verifying that the reduced form is economically coherent, as recommended by Angrist and Pischke (2008), and providing the inputs for the Plausibly Exogenous analysis that follows. The reduced form estimates are in line with prior expectations and are used to calibrate the bounds $\gamma_0 = \pm 0.5\gamma$ for the sensitivity exercise.

4) Plausibly Exogenous Bounds (Panel E): The final panel reports 95% union-of-confidence-intervals bounds for the instrumented coefficient $\hat{\gamma}$, relaxing the strict exogeneity assumption underlying the exclusion restriction. In all four cases, the confidence intervals exclude zero by a wide margin, reinforcing confidence in the 2SLS estimates. Furthermore, the intervals are asymmetric relative to the OLS coefficient, indicating that allowing for a direct effect of the instrument on the outcome generally strengthens - rather than undermines - the estimated positive effect of internet availability on expenditure.

The confidence intervals for $\hat{\gamma}$ derived under the plausible exogeneity assumption. Please see Figs. B1-B4 for robustness checks.

5) Robustness: All regressions are reproduced in levels. Given the right-skewness of the consumption distribution - and the concern that results could be driven by very wealthy households at the upper tail - we additionally report estimates after trimming the sample by 2% (1% at each tail) and 10% (5% at each tail) to confirm robustness to extreme values.

5. Matching

Our objective with matching is to address confounding factors and selection bias through an approach that is conceptually distinct from 2SLS.

Table 3 reports the balance of observable characteristics between households with and without internet access before and after applying Coarsened Exact Matching (CEM). The picture that emerges is one of strong positive selection into internet access. Before matching, *t*-tests of differences in means and proportions are significant across all covariates, with more educated and wealthier households systematically enjoying greater access. Gradients across education levels and wealth quintiles are particularly pronounced. This positive selection is partially offset by two variables - 'Other Backward Class' and 'Rural' - whose coefficients indicate significantly higher internet access, introducing a degree of negative selection that somewhat tempers the overall pattern.

After matching, these differences shrink dramatically. Only 'household size' remains statistically significant, attesting to the strong performance of the CEM algorithm. Overall, the matching procedure achieves near-perfect covariate balance, substantially reducing pre-existing differences between treated and control households. The matched sample thus approximates a setting in which treatment assignment is independent of observed characteristics, strengthening the credibility of the causal estimates that follow in Table 4.

Households with internet access tend to be wealthier, better educated, and concentrated in more economically dynamic areas - characteristics that independently drive higher consumption. Because OLS does not account for this selection, its coefficients absorb both the effect of internet access and the confounding influence of these correlated predictors, likely inflating the estimated magnitude.

Coarsened Exact Matching addresses this by restricting estimation to the region of common support: observations are matched on observable characteristics, and unmatched units are discarded. OLS, by contrast, uses the full sample and weights all observations equally, which means it implicitly extrapolates into regions where treated and control units do not overlap. This extrapolation is a second source of upward bias in the OLS estimate.

Table 4 reports estimates obtained via Coarsened Exact Matching (CEM). As anticipated given the positive selection characterizing our sample, the coefficients of interest are smaller in magnitude under CEM than under OLS - and *a fortiori* under 2SLS. A difference of $\hat{\beta}$

Table 4
Coarsened exact matching: internet availability and expenditure (logarithm).

	LN of Total Expenditure	LN of Food Expenditure	LN of Consumables Expenditure	LN of Durables Expenditure
β FSU FE	.178***	.079***	.22***	.171***
(SE)	(0.006)	(0.004)	(0.007)	(0.006)
R2	.587	.64	.514	.557
β District FE	.163***	.077***	.195***	.166***
(SE)	(0.006)	(0.005)	(0.008)	(0.007)
R2	.41	.382	.34	.332
N	239,695	239,112	239,278	237,215

Source: Author's Calculation.

emerges between the FSU and district fixed effects specifications, which we attribute to more pronounced attenuation bias in the district fixed effects case.

We observe the following ordering among the three estimators:

$$\widehat{\gamma}_{CEM} < \widehat{\gamma}_{OLS} < \widehat{\gamma}_{2SLS}$$

This ranking is consistent with our earlier discussion and can be understood through two complementary lenses.

First, OLS is subject to attenuation bias in the presence of classical measurement error in the regressor - it systematically underestimates the true coefficient magnitude. CEM, by contrast, uses a coarser version of the treatment variable, which amplifies this attenuation, pushing further downward. The 2SLS estimator, by instrumenting for the mismeasured regressor, recovers a consistent estimate, yielding the largest of the three. Second, the logarithmic transformation of the outcome interacts with these biases in a predictable way: the logarithm is a concave function, it symmetrises the outcome distribution and modifies the estimate which becomes the proportional treatment effect. These features of the logarithm reinforce the ordering we observe. Please see [Tables C1–C3](#) for additional details on matching.

[Table 5](#) presents regressions of internet availability on diet components, comparing FSU and District fixed effects using both OLS and CEM estimators. The divergence between these two fixed-effects specifications reveals that diet components tied to fresh food are highly sensitive to household location - outside major metropolitan areas, fresh food cannot be ordered or delivered online. We therefore conclude that District fixed effects are insufficient in this context, as they fail to absorb enough unobserved heterogeneity. Consequently, we do not report 2SLS estimates, since our instrument loses validity under FSU fixed effects.

Taken together, this empirical exercise - drawing on multiple identification strategies across a range of dependent variables - robustly establishes that internet availability increases both household expenditures and consumption.

6. Conclusions and policy recommendations

This study adds to the growing body of research from various developing countries looking into how access to the internet is a powerful driver of household consumption, diversity and dietary components of it. Access to internet leads to better market awareness and facilitates e-commerce services ([Cui et al., 2023](#); [Sentosa et al., 2025](#); [Gao and Li, 2024](#); [Xue et al., 2021](#); [Baojie and Jin, 2022](#); [Yang et al., 2023](#); [Deng et al., 2024](#)). Findings from this study are well aligned with that from rural China, where internet access has led to improve dietary quality and overall food consumption through increased income opportunities and access to variety of foods ([Cui et al., 2023](#); [Xue et al., 2021](#); [Baojie and Jin, 2022](#); [Yang et al., 2023](#)). However, it is also pertinent to note that in other settings, these benefits are not evenly distributed: socioeconomic status, education, and social identity mediate the extent to which households can translate digital access into improved welfare and nutrition ([Agrawal et al., 2019](#); [Deng et al., 2024](#); [Padmaja et al., 2019](#)).

Findings have suggested that internet access exerts a substantial influence on household consumption behaviour in India. Households with digital connectivity report higher overall expenditure, particularly on durable goods, consumables, and discretionary items. This pattern reflects the role of the internet in expanding market awareness, facilitating e-commerce, and enabling digital financial services. Importantly, the effects are not limited to expenditure; internet use also reshapes dietary choices. Connected households exhibit higher caloric intake and greater consumption of proteins, carbohydrates, and fats, alongside a broader diversity of food consumption. These findings suggest that digital connectivity expands dietary options through exposure to diverse cuisines, online food delivery platforms, and nutritional resources, while also posing potential risks associated with unhealthy eating patterns.

These findings underscore the need for targeted policy interventions that not only expand internet access but also ensure that digital connectivity translates into meaningful welfare improvements. First, bridging the digital divide requires investments in broadband infrastructure and subsidies to make internet access affordable, particularly in rural and marginalized communities.

Table 5

Coarsened exact matching: internet availability and diet (logarithm).

	LN of Total Daily Calories	LN of Total Daily Proteins	LN of Total Daily Carbohydrates	LN of Total Daily Fats
OLS				
β FSU FE	.087***	.084***	.085***	.086***
(SE)	(0.002)	(0.003)	(0.003)	(0.004)
R2	.724	.779	.783	.675
β District FE	.049***	.02***	.031***	.051***
(SE)	(0.003)	(0.004)	(0.005)	(0.005)
R2	.567	.514	.575	.423
N	257,936	258,021	261,064	261,066
CE Matching				
β FSU FE	.045***	.035***	.04***	.044***
(SE)	(0.003)	(0.003)	(0.003)	(0.004)
R2	.698	.777	.782	.666
β District FE	.013***	−0.025***	−0.007	.039***
(SE)	(0.003)	(0.005)	(0.005)	(0.005)
R2	.504	.471	.548	.379
N	236,236	236,318	239,102	239,103

Source: Author's Calculation.

Second, digital literacy initiatives should be integrated with nutrition education and financial inclusion programs to promote healthier food choices and enable households to use digital platforms effectively. Third, tailored interventions for vulnerable groups -such as women-led households, lower-income families, and marginalized castes -are necessary to ensure that internet access reduces rather than reinforces existing inequalities. Finally, integrating e-commerce and fintech innovations with social protection schemes can enhance food security, diversify consumption, and create more resilient household economies.

Finally, by situating these findings within the global literature, this study underpins the need for policy interventions that go beyond infrastructure expansion to address digital literacy, nutrition education, and targeted support for vulnerable groups (Cui et al., 2023; Sentosa et al., 2025; Deng et al., 2024; Rahman et al., 2025). Integrating digital tools with social protection, financial inclusion, and regionally tailored nutrition programs can help ensure that the benefits of connectivity are equitably realized. Conclusively, this study demonstrates that while access to the internet is a catalyst for changing household consumption and dietary patterns in India, its transformative potential for inclusive and sustainable development depends on policies that address the complex interplay of digital connectivity, social landscape, and economic factors.

7. Limitations and directions for future research

While the analysis provides consistent evidence of a positive effect of internet access on household expenditure, several limitations should be acknowledged. First, despite the use of an instrumental variables strategy and complementary matching techniques, the validity of the causal interpretation ultimately depends on the strength of the exclusion restriction. Although robustness checks, including the plausibly exogenous framework, suggest that the results are not sensitive to moderate violations, the instrument -based on local internet penetration may still capture unobserved factors correlated with consumption. Second, the analysis relies on cross-sectional data, which limits the ability to control for time-invariant unobserved heterogeneity and to trace dynamic effects of internet adoption over time. Third, the empirical strategy does not allow for a clear disentangling of underlying mechanisms. In particular, the estimated effects may reflect a combination of improved access to information (e.g., prices, employment opportunities) and expanded access to markets through digital platforms (e-commerce), which cannot be separately identified in the current framework. The availability of multiple rounds of the Household Consumption and Expenditure Survey (HCES) also opens the possibility of constructing a district-level panel, which would allow the use of panel data methods to better control for unobserved heterogeneity and strengthen causal identification.

These limitations point to several avenues for future research. A key direction would be to distinguish between the “information channel” and the “purchase channel” through which internet access affects household behavior, potentially using more granular data on online activities or transaction-level information. Furthermore, future studies could strengthen causal identification by exploiting quasi-experimental variation in internet availability, such as the staggered rollout of broadband infrastructure, policy-driven expansions in digital connectivity, or geographic variation in network coverage. The use of panel data would also allow researchers to better account for unobserved heterogeneity and to examine longer-term impacts of internet adoption on consumption patterns. Together, such approaches would provide a deeper understanding of both the mechanisms and the persistence of the effects documented in this study.

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CRediT authorship contribution statement

B Sai Swaroop: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Olivier Dagnelie:** Formal analysis, Methodology, Validation, Writing – review & editing. **Rajabhushan Jagadish Nayak:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Akanksha Choudhary:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Ashish Singh:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Definition of our variables

The **dependent variables** are derived from multiple modules of the Household Consumption and Expenditure Survey, including Food Consumption (FDQ), Consumables and Services (CSQ), and Durable Goods (DGQ).

● Total Household Expenditure

Total expenditure is computed by aggregating expenditures across all consumption categories. Specifically:

- **Food expenditure** is derived from Levels 05 and 06 (FDQ), which report item-wise consumption quantities and values (in Rs.) over the reference period.
- **Consumables and services expenditure** is obtained from Levels 08, 09, and 10 (CSQ), which include non-food recurring expenditures such as fuel, education, health, and other services.
- **Durable goods expenditure** is computed from Levels 11–13 (DGQ), which capture expenditures on household assets, including both first-hand and second-hand purchases, as well as repair and maintenance costs.
- These components are aggregated to obtain **total household expenditure (in Indian Rupees)**.

● Macronutrient Intake values

Nutritional variables—**calories, proteins, fats, and carbohydrates**—are computed using food consumption quantities from FDQ modules. These quantities are converted into nutrient values using standard conversion factors (based on USDA food composition tables) and aggregated at the household level.

For example, if a household consumes a given quantity of rice, the corresponding carbohydrate, protein, fat, and calorie content is calculated using standard nutritional values for rice, and this is repeated for all food items before aggregating to the household level.

● Social Category Variables

● **Scheduled Caste (SC)**: Indicator equal to 1 if the individual belongs to a Scheduled Caste—groups historically subjected to social exclusion within the caste system—and 0 otherwise.

● **Scheduled Tribe (ST)**: Indicator equal to 1 if the individual belongs to a Scheduled Tribe, referring to constitutionally recognized indigenous communities with distinct socio-cultural identities and historically limited access to mainstream economic development, and 0 otherwise.

● **Other Backward Class (OBC)**: Indicator equal to 1 if the individual belongs to Other Backward Classes, a category of socially and educationally disadvantaged groups identified for affirmative action, and 0 otherwise.

Appendix B. Robustness checks

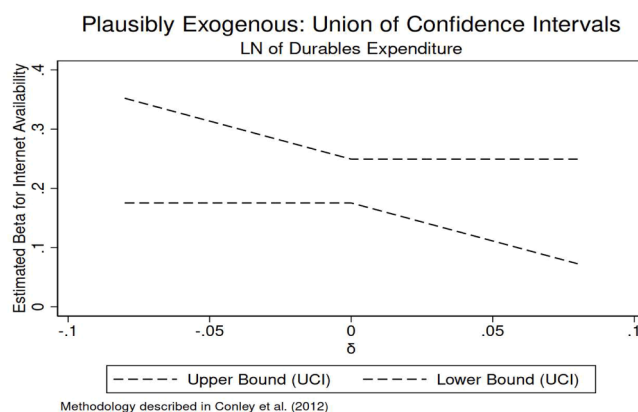


Fig. B1. Plausibly Exogenous Confidence Intervals for Internet Access (Durables Expenditure)

Notes: This figure reports the union of confidence intervals (UCI) for the estimated effect of internet access on the logarithm of durables expenditure under the plausibly exogenous framework. The horizontal axis represents the assumed direct effect of the instrument (δ), while the vertical axis shows the corresponding estimated coefficient on internet access. The dashed lines denote the upper and lower bounds of the confidence interval. Results are robust to moderate deviations from the exclusion restriction, following the methodology of [Conley et al. \(2012\)](#).

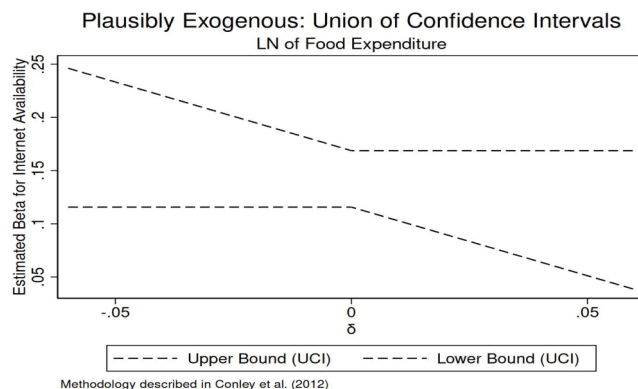
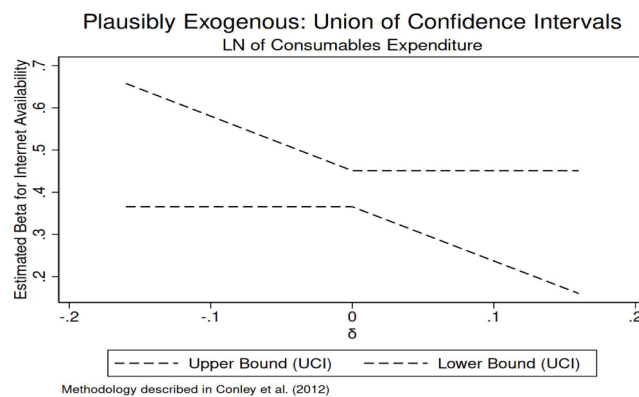
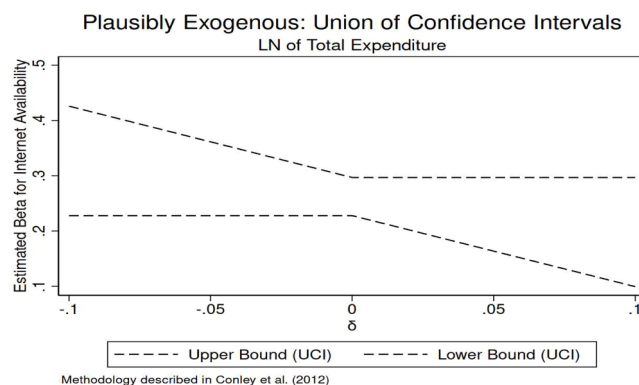


Fig. B2. Plausibly Exogenous Confidence Intervals for Internet Access (Food Expenditure)

Notes: This figure reports the union of confidence intervals (UCI) for the estimated effect of internet access on the logarithm of Food expenditure under the plausibly exogenous framework. The horizontal axis represents the assumed direct effect of the instrument (δ), while the vertical axis shows the corresponding estimated coefficient on internet access. The dashed lines denote the upper and lower bounds of the confidence interval. Results are robust to moderate deviations from the exclusion restriction, following the methodology of Conley et al. (2012).

**Fig. B3.** Plausibly Exogenous Confidence Intervals for Internet Access (Consumables Expenditure)

Notes: This figure reports the union of confidence intervals (UCI) for the estimated effect of internet access on the logarithm of Consumables expenditure under the plausibly exogenous framework. The horizontal axis represents the assumed direct effect of the instrument (δ), while the vertical axis shows the corresponding estimated coefficient on internet access. The dashed lines denote the upper and lower bounds of the confidence interval. Results are robust to moderate deviations from the exclusion restriction, following the methodology of Conley et al. (2012).

**Fig. B4.** Plausibly Exogenous Confidence Intervals for Internet Access (Total Expenditure)

Notes: This figure reports the union of confidence intervals (UCI) for the estimated effect of internet access on the logarithm of Total expenditure under the plausibly exogenous framework. The horizontal axis represents the assumed direct effect of the instrument (δ), while the vertical axis shows the corresponding estimated coefficient on internet access. The dashed lines denote the upper and lower bounds of the confidence interval. Results are robust to moderate deviations from the exclusion restriction, following the methodology of Conley et al. (2012).

Appendix C. Matching tables

Table C1

Coarsened exact matching: internet availability and expenditure.

CE Matching	Total Expenditure	Food Expenditure	Consumables Expenditure	Durables Expenditure
β FSU FE	8475.123***	226.971***	4217.462***	4687.503***
(SE)	(593.482)	(21.198)	(360.086)	(435.428)
R2	.315	.459	.262	.242
β District FE	8192.685***	262.371***	4164.227***	4535.196***
(SE)	(641.756)	(25.552)	(341.778)	(489.02)
R2	.178	.201	.128	.106
N	239,695	239,112	239,278	237,215

Table C2

Internet availability and expenditure.

OLS	Total Expenditure	Food Expenditure	Consumables Expenditure	Durables Expenditure
β FSU FE	6675.561***	174.436***	2781.612***	4170.52***
(SE)	(412.458)	(14.675)	(268.14)	(301.002)
R2	.32	.476	.255	.238
β District FE	6836.87***	209.065***	3061.883***	4046.436***
(SE)	(410.403)	(16.674)	(247.252)	(288.593)
R2	.205	.253	.141	.121
N	261,739	261,075	261,257	258,995
Control function: District FE				
β	9580.825***	561.617***	6695.454***	3444.373***
(SE)	(1248.281)	(57.043)	(719.198)	(870.071)
1st st residuals	−3462.429**	−443.285***	−4584.339***	759.465
(SE)	(1345.355)	(59.596)	(790.16)	(940.345)
N	261,729	261,065	261,247	258,985
R2	.205	.254	.142	.121
2SLS: District FE				
β	9580.825***	562.346***	6699.03***	3445.757***
(SE)	(1249.617)	(57.126)	(721.219)	(868.223)
N	261,729	261,065	261,247	258,985
R2	.191	.232	.126	.109
Reduced form: District FE				
β	7441.772***	436.077***	5199.322***	2681.332***
(SE)	(968.348)	(44.326)	(558.379)	(675.035)
N	261,729	261,065	261,247	258,985
R2	.204	.254	.141	.12
Plausibly exogenous: UCI: District FE				
95% CI of β	[7135.046; 16,820.519]	[450.525; 957.568]	[5286.062; 11,459.455]	[1743.622; 6868.609]
N	261,729	261,065	261,247	258,985

Table C3

Coarsened exact matching: internet availability and diet.

OLS	Total Daily Calories	Total Daily Proteins	Total Daily Carbohydrates	Total Daily Fats
β FSU FE	186.976***	.824**	18.82***	5.979***
(SE)	(24.992)	(0.332)	(2.278)	(0.687)
R2	.393	.758	.772	.455
β District FE	23.385	−3.524***	−8.951***	4.786***
(SE)	(29.956)	(0.498)	(3.215)	(0.85)
R2	.264	.517	.588	.252
N	257,936	258,036	261,075	261,075
CE Matching				
β FSU FE	207.629***	.641*	16.003***	4.761***
(SE)	(31.699)	(0.376)	(2.608)	(0.819)
R2	.348	.742	.756	.442
β District FE	45.57	−3.806***	−12.851***	6.92***
(SE)	(35.17)	(0.593)	(3.832)	(0.991)
R2	.199	.453	.525	.207
N	236,236	236,329	239,112	239,112

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